Advanced AI Models in Engineering

Project Topic and Task Split

Topic (Theme 5.1): Optimizing an RL pendulum-balancing policy for memory footprint. An RL agent learns to balance a pendulum in a digital twin. An optimization framework then searches for the best trade-off between balancing performance and memory footprint, with the full agent as the baseline.

Task	Student A	Student B
1–6 Planning	Research the project	Research the project
7–8	Research and build the digital twin	Research Gym and the optimization framework
9	Set up the RL model to learn balancing	Set up the optimization: keep baseline success while reducing memory footprint, find which parts of the model are worth keeping
10–11	Prepare visualizations and conclusions	Prepare visualizations and conclusions
12	Defense presentation	Defense presentation